

Elements of Statistics and Probability

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1 What is Statistics?

What is Statistics?

- In ancient times, Statistics was known as the “Science of Statecraft” and emerged as a by-product of governmental administrative activities.
- Origin of the word ‘Statistics’
 - Latin word Status
 - Italian word Statista
 - German word Statistik
 - French word Statistique
- Each of which means a **political state**

“Prophetess” in the Development of App. Stat.



Florence Nightingale

- Unknown to many, Florence Nightingale used new techniques of statistical analysis
- During the Crimean War, she plotted the incidence of preventable deaths in the military
- She developed the “polar-area diagram” to highlight the needless deaths caused by unsanitary conditions and the need for reform
- Her analysis revolutionized the idea that social phenomena could be objectively measured and subjected to mathematical analysis

Polar-Area Diagram

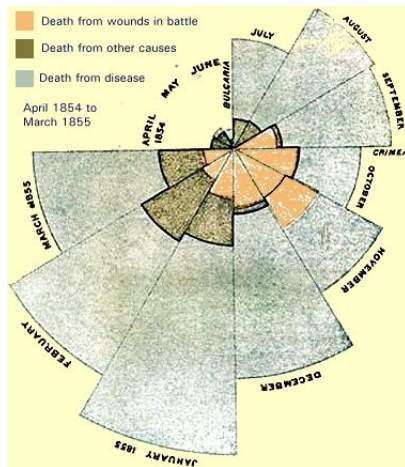


Figure 1: Polar-area diagram by Florence Nightingale

Definition of Statistics

- According to Sir R.A. Fisher ([1947](#)), the science of statistics is essentially a branch of applied mathematics and may be regarded as mathematics, applied to observational data.
- Statistics is a branch of mathematics dealing with the collection, analysis, interpretation, and presentation of masses of numerical data ([Merriam Webster](#))
- **Statistics** is the science of collecting, organizing, presenting, analyzing, and interpreting numerical data for the purpose of assisting in making a more effective decision

Objectives of Statistics

- The main objectives of statistics are
 - Collecting the data
 - Summarizing the data
 - Estimating parameters from the data
 - Hypothesis testing
 - Interpreting the results
 - Drawing valid conclusion
 - Forecasting the future values

Example

- Let's consider an example that involves using statistics to analyze CGPA (Cumulative Grade Point Average) in a university setting

Example (cont.)

- **Collecting the data:** CGPA data are collected from the students, including information such as the student's name, department, semester etc
- **Summarizing the data:** The collected CGPA data is summarized and organized.(a table, or graph)
- **Estimating parameters from the data:** Statistical techniques can be used to estimate parameters related to the CGPA data. For example, the average CGPA of all students can be estimates

Example (cont.)

- **Testing the hypothesis:** We may have a hypothesis, such as “the average CGPA of engineering students is higher than that of business students”
- **Interpreting the results:** The results of the statistical analysis are interpreted in the context of the CGPA data
- **Drawing valid conclusions:** We may conclude that engineering students have a significantly higher average CGPA compared to business students
- **Forecasting the future values:** We can forecast the average CGPA for the upcoming semester

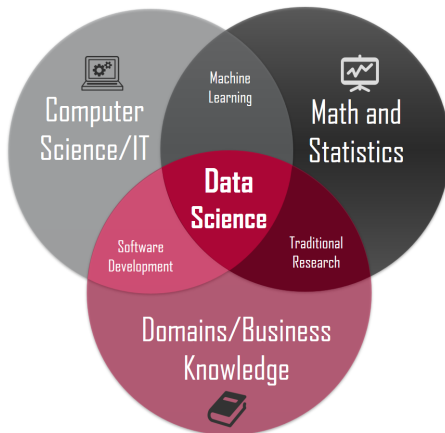
Applications of Statistics

- Insurance companies use statistics to set rates for different insurance
- City corporation may want to examine the contamination level of different lake water
- Medical professional assess the performance of a new drug over the existing one
- No matter what is your future line of work, you will make decisions that will involve data

Scopes of Statistics

- **Science:** Computer science, Medicine, Genetics, Epidemiology, Environmental studies, Astronomy
- **Arts and social science:** Economics, Sociology, Politics, Psychology, Development studies
- **Business:** Finance, Banking, Insurance, Actuary, Marketing

Statistics and Data Science



Statistics and Data Science

- Data science comprises three distinct and overlapping areas:
 - the skills of a **statistician** who knows how to model and summarize datasets
 - the skills of a **computer scientist** who can design and use algorithms to efficiently store, process, and visualize this data
 - the **domain expertise** — what we might think of as **classical** training in a subject— necessary both to formulate the right questions and to put their answers in context

Types of Statistics

- **Descriptive Statistics:** Methods of organizing, summarizing, and presenting data in an informative way
 - **Measures of central tendency:** mean, median, mode
 - **Measures of dispersion:** range, variance, standard deviation
 - **Example:** A survey in a secondary school found that 49% of the students knew writer Humayun Ahmed. The statistic 49 describes the number out of every 100 persons who knew the answer

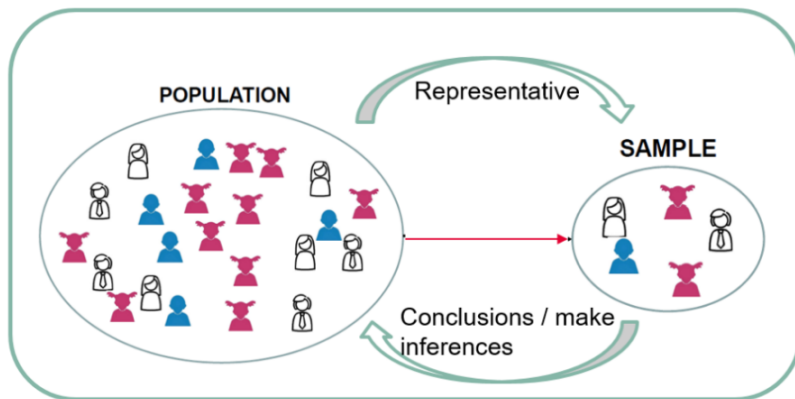
Types of Statistics (cont.)

- **Inferential Statistics:** Involves making assumptions/drawing conclusions about a **population** based on a **sample**
 - **Hypothesis testing:** z test, t test, ANOVA
 - **Regression analysis:** linear regression, binary logistic regression
 - **Example:** The accounting department of Aarong will select a sample of the invoices to check for accuracy for all the invoices of Aarong

Population and Sample

- **Population:** a collection of all possible individuals, objects, or measurements of interest (with specific characteristics)
- **Sample:** a representative portion or part of the **population** of interest
- **Example:** To determine the average height of the newly enrolled students at BracU, data on heights have been collected from 1st semester students of the CSE and Economics department
 - population: ?
 - sample: ?

Population and Sample (cont.)



Parameter and Statistic

- **Parameter:** a characteristic or measure (numerical) obtained by using data values from a specific **population**
- **Statistic:** a characteristic or measure (numerical) obtained by using the data values from a **sample**
- **Example:** To determine the average height of the newly enrolled students at BracU, data on heights have been collected from 1st semester students of the CSE and Economics department
 - **parameter:** ?
 - **sample:** ?

Variable

- **Variable:** a characteristic of interest for the individual which takes on different values in different individual. For example, age, gender, height, weight, eye color, wealth status
 - **Qualitative (attribute):** a non-numeric valued variable.
Example: gender (male, female), type of residence (urban, rural)
 - **Quantitative:** a variable that can be measured and reported numerically. **Example:** individual's BMI, time remaining in a class, age at first marriage

Quantitative Variable

- **Discrete:** can assume only isolated values within a given range.
Example: number of students in a class, number of dengue cases in a year, number of courses in a semester
- **Continuous:** can take any values within a specific range.
Example: age of individuals, birth weight of children, CGPA of students

Scales of Measurement

- **Measurement** means assigning numbers or other symbols to characteristics of objects according to certain prescribed rules
- **Scaling** is the generation of a continuum upon which measured objects are located
- Statistical operations for summarizing and presenting data depend on the levels of measurement

Scales of Measurement (cont.)

- **Four** scales of measurement
 - **Nominal**: data that can only be classified into categories and cannot be arranged in an ordering scheme
 - **Ordinal**: involves data that may be arranged in some order, but differences between data values cannot be determined or are meaningless
 - **Interval**: numerical data that arithmetic operations are meaningful
 - **Ratio**: differences and ratios are meaningful for this level of measurement

Nominal Scale

- **Example:** eye color, gender, religious affiliation
- Categories must be **mutually exclusive** and **exhaustive**
 - **Mutually exclusive:** categories are so defined that each member of the population is correctly allocated to one and only one category
 - **Exhaustive:** each person, object, or item must be classified in at least one category, i.e., classification is exhaustive when it provides sufficient categories to accommodate all members of the population

Ordinal Scale

- Numeric codes are used to identify the different categories of a variable in such a way that these numbers or values can be arranged in ascending or descending order
- **Example:** socioeconomic status, level of education
- Categories are **mutually exclusive** and **exhaustive**
- Differences between the category 1 and category 2, and category 2 and category 3 are not the same

Interval Scale

- Similar to the ordinal level, with the additional property that meaningful amounts of differences between data values can be determined
- There is no natural zero point
- **Example:** temperature in centigrade
- 0°C does not mean the 'no temperature!'

Ratio Scale

- Interval level with an inherent zero starting point
- Zero means absence of the characteristics
- Differences and ratios are meaningful for this level of measurement
- **Example:** age, weight, pulse rate, money

Scales of Measurement (cont.)

Scale	Characteristic Question	Examples
Nominal	Is A different than B?	Marital status, Religious affiliation, Race
Ordinal	Is A bigger than B?	Severity of pain, Level of satisfaction
Interval	By how many units do A and B differ?	Temperature, SAT score
Ratio	How many times bigger than B is A?	Distance, Length, Time until death

Scales of Measurement

Scale	Operation that make sense			
	Counting	Ranking	Addition/subtraction	Division/multiplication
Nominal	✓			
Ordinal	✓	✓		
Interval	✓	✓	✓	
Ratio	✓	✓	✓	✓

- Often, the distinction between interval and ratio scales can be ignored in statistical analyses. Distinction between these two types and ordinal and nominal are more important.

Sources of Data

- **Data:** Information obtained by observing values of a variable
- Data on these research problems may be found in published articles, journals, and magazines (**secondary data**)
- If published data is not available on a given subject. In such cases, information have to be collected (**primary data**)
- There are different methods of collecting data

Sources of Data (cont.)

- **Primary data** are originated by a researcher for the specific purpose of addressing the research problem at hand. It is expensive and time consuming
- **Secondary data** are data that have already been collected for purposes other than the research problem at hand. It is inexpensive and fast
- **Example:** Price of rice in Dhaka city market. Researcher may collect data from the retailers directly the data obtained this way is **primary data**. He/She may use data published in different daily news papers, then it will be **secondary data**

Sources of Data

- Data collection instruments (**Questionnaire**/**Schedule**): formalized set of questions for obtaining information from respondents. **Questionnaire** is filled by **respondent** and **Schedule** is filled by the **interviewer**
- Observation
- Personal interview
- Telephone Conversation
- Mailing
- Combination

Objectives of a Questionnaire/Schedule

- A questionnaire must translate the information needed into a set of specific questions that the respondents can and will answer
- It must motivate and encourage the respondent to become involved in the interview, to cooperate, and to complete the interview
- A questionnaire should minimize the response error

Processing of Data

- Checking
- Editing
- Coding
- Data entry
- Cleaning
- Analyzing

Exercise 1

1. Slate is a daily magazine on the Web. Its business activities can be described by a number of variables. What is the level of measurement for each of the following variables?
 - (a) The number of hits on their website on Saturday between 8:00 am and 9:00 am
 - (b) The departments, such as food and drink, politics, foreign policy, sports, etc
 - (c) The number of weekly hits on the Sam's Club ad.
 - (d) The number of years each employee has been employed with Slate

Exercise 2

2. For each of the following, determine whether the group is a sample or a population
- (a) The participants in a study of a new cholesterol drug
 - (b) The drivers who received a speeding ticket in Kansas City last month
 - (c) People on welfare in Cook County (Chicago), Illinois
 - (d) The 30 stocks that make up the Dow Jones Industrial Average

Exercise 3

3. Place these variables in the following classification tables. For each table, summarize your observations and evaluate if the results are generally true. For example, salary is reported as a continuous quantitative variable. It is also a continuous ratio-scaled variable.
- salary, gender, sales volume of MP3 players, soft drink preference, temperature, SAT scores, student rank in class, rating of a CSE professor, number of home video screens

Exercise 3 (cont.)

	Discrete Variable	Continuous Variable
Qualitative		
Quantitative		salary,...

	Discrete Variable	Continuous Variable
Nominal		
Ordinal		
Interval		
Ratio		salary,...

Exercise 4

4. Suppose a statistician wants to analyze the data of a particular telecom company. For this purpose, he collected data from the server of this company. He collected data on Gender, hours internet browsing per day, number of SMS sent per day, hours spent on incoming or outgoing call, brand of mobile phone (iPhone, One plus, Samsung, Xiaomi, Huawei, Oppo etc.), User Status (Platinum, Gold and Silver), age of the customer and duration of using that particular brand (e.g. 2 years, 1.5 years or 3.11 years) etc. Now find the following things:

Exercise 4 (cont.)

- (a) Sample and Population
- (b) Identify qualitative and quantitative variables
- (c) Identify the scale of measurement for each variable (nominal, ordinal, interval or ratio)
- (d) Identify discrete or continuous variables in case of continuous variables

Thank You